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Optimization of apparent polar wander paths: An example from the South China plate

Stuart A. Gilder[a,*], Xiaodong Tan[b], Hugo Bucher[c], Guodun Kuang[d], Jiarun Yin[e]

[a] Ludwig Maximilians Universität, Department of Earth and Environmental Sciences, Theresienstrasse 41, 80333 Munich, Germany
[b] Chinese Academy of Sciences, Key Laboratory of Marginal Sea Geology, South China Sea Institute of Oceanology, 164 West Xin Gang Road, 510301 Guangzhou, China
[c] Paläontologisches Institut der Universität Zürich, Karl Schmid-Strasse 4, 8006 Zürich, Switzerland
[d] Guangxi Bureau of Geology and Mineral Resources, Jiangzheng Road 1, 530023 Nanning, China
[e] China University of Geosciences, Xueyuan Road 29, 100083 Beijing, China

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ABSTRACT

Paleomagnetically derived apparent polar wander paths form the foundation of quantitative plate tectonic reconstructions. However, deformation leading to vertical axis block rotations displaces paleomagnetic poles away from their original positions, leading to an ambiguity as to which pole, or group of poles, best approximates the "true" reference pole position for a given time. Here we show that the best estimate of the "true" reference pole will match the observed paleolatitude (pλ) for each point on the plate. This means that the expected pλ from the "true" reference pole minus the observed pλ, derived from each individual study, will average to zero. Histogram plots and associated parameters help further discriminate between candidate reference poles when more than one of them fulfills the zero-average requirement within prescribed uncertainty limits. Our analysis of 44 Late Permian to Middle Triassic paleomagnetic poles from the South China plate corroborates previous assumptions that the poles from Sichuan Province best represent the "true" reference for the South China plate. To better understand the age of the rotations, we studied the paleomagnetism of Lower and Upper Triassic rocks from the Shiwandashan region in Guangxi Province. Early Triassic paleomagnetic directions isolated at high temperature demagnetization steps are of dual polarity and pass the fold test. This magnetization component is indistinguishable at 95% confidence limits from Middle Triassic paleomagnetic directions from other parts of Guangxi Province. The corresponding pole for this component lies within the swath of Early to Middle Triassic paleomagnetic poles from the South China plate, confirming that Guangxi has been a part of the South China plate since at least the Early Triassic. Late Triassic paleomagnetic data require further study before their complex magnetizations can be interpreted.

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1. Introduction

Understanding how plates move and representing those motions on paleogeographic maps constitute an important goal for paleomagnetists. To accomplish this, a quasi-continuous time sequence of paleomagnetic poles must be obtained to construct an apparent polar wander path (APWP) for each plate in order to constrain their relative motions through time. However, defining an APWP is not trivial, as plates might not remain rigid over pluri-million year time scales. Non-rigidity can cause paleomagnetic poles to rotate away from their original positions, which makes the APWP ambiguous, and thus renders tectonic reconstructions subject to error.

South China represents one such case where Lower Permian to Middle Triassic rocks display highly discordant paleomagnetic data (Fig. 1). In the plate's northern and western parts, declinations are northeast-directed where fold axes trend northeast, whereas declinations are east-directed where fold axes trend eastward (Huang and Opdyke, 1996, 1997; Yang and Besse, 2001; Su et al., 2005; Tan et al., 2007a). This general clockwise tendency of both the rotations and the fold axes led workers to examine the process that created the pattern under the assumption that the east-directed fold axes and concomitant rotations were emplaced during deformation related to the collision between the North and South China plates (see Fig. 15 of Gilder et al., 1999, for a synthesis of the Mesozoic to Present tectonic evolution of east China).

Assuming the North and South China plate collision produced the paleomagnetic rotations and the folding and faulting in the vicinity surrounding the suture zone, one must explain why Lower to Middle Triassic rocks about 1500 km away in Guangxi Province

* Corresponding author.
E-mail address: gilder@lmu.de (S.A. Gilder).

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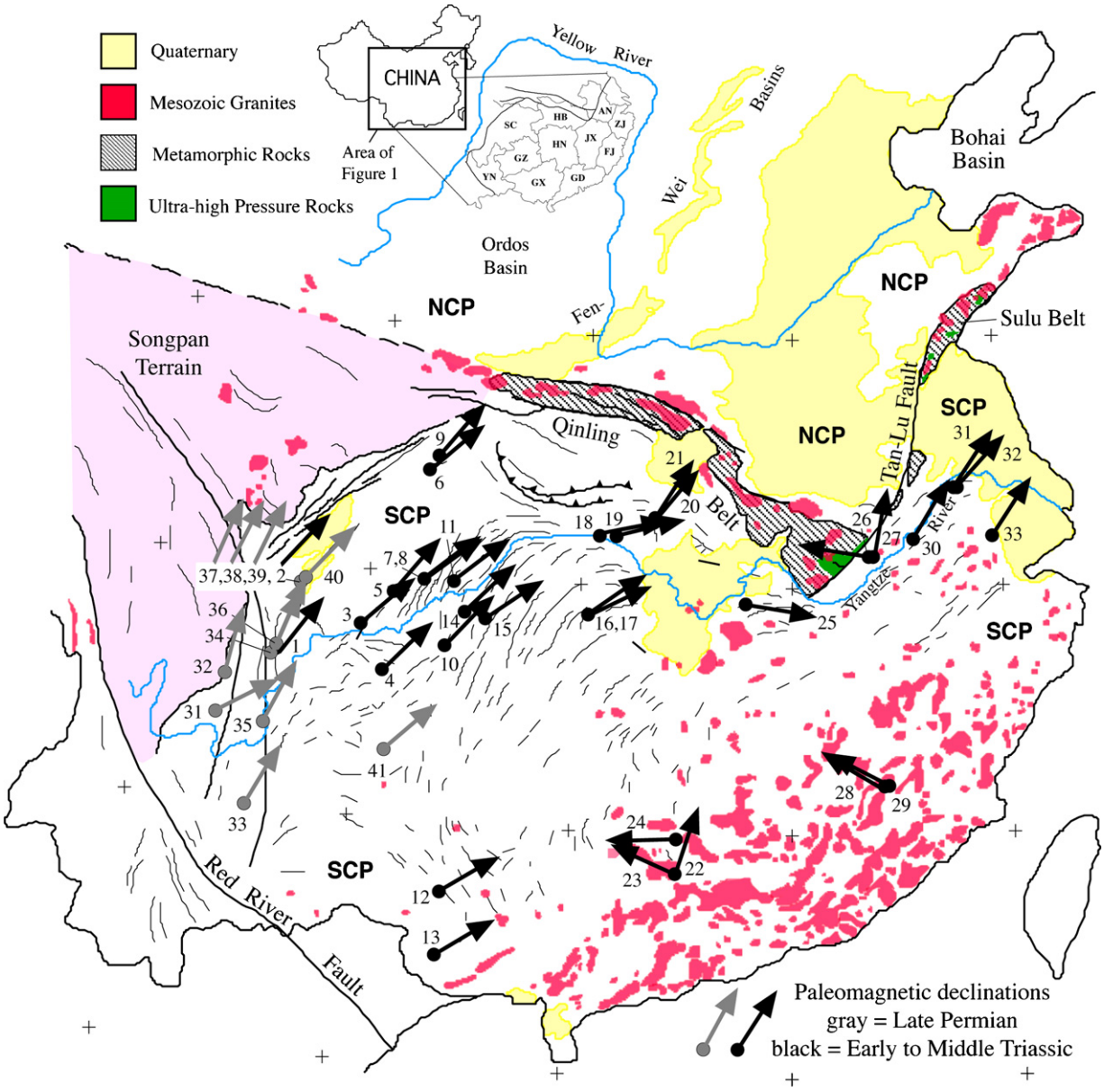


Fig. 1. Simplified geological map of the South China plate (SCP) (after Academia Sinica, 1976) showing Late Permian to Middle Triassic paleomagnetic declinations. The South China plate is defined as the region east of the Songpan Terrain (in light pink), east of the Red River and Tan-Lu faults, and south of the Qinling and Sulu belts. The latter two form the suture between the SCP and the North China plate (NCP). Numbers next to paleomagnetic localities correspond to studies in Table 2. Provinces in the inset are abbreviated as follows: AN, Anhui; FJ, Fujian; GD, Guangdong; GX, Guangxi; GZ, Guizhou; HB, Hubei; HN, Hunan; JX, Jiangxi; SC, Sichuan; YN, Yunnan; ZJ, Zhejiang.

possess paleomagnetic rotations similar to those near the suture (Fig. 1). One could also question whether the declinations in Sichuan were rotated counterclockwise during the India–Asia collision as South China subducts under the Songpan terrain near the Longmenshan Mountains. In other words, the "true" declinations representative of the South China plate could be those near the Qinling suture, while those from Sichuan rotated counterclockwise away from them. Or perhaps various deformation events have deviated all of the South China Upper Permian to Middle Triassic poles away from their original positions. Moreover, Van der Voo et al. (1993) noted a north–northeast versus east–northeast dichotomy of paleomagnetic declinations in the Permian Emeishan basalts. They found both north and northeast-pointing declinations in the same section. Electron microscopy determinations together with rock magnetic experiments led them to conclude that both groups of directions spawned from primary magnetizations. This raises the question whether the different directional sets are instead related to a geomagnetic field phenomenon.

Typically Sichuan Province has been assumed to represent "stable" South China so the average Sichuan pole has been considered the reference with which to calculate rotation magnitudes for the rest of the plate. However, since the Permo-Triassic poles are streaked out about a swath exceeding 90° (Fig. 2), one does not know which pole correctly represents the South China plate. This is important because unraveling the tectonic history of South China depends upon knowing the "true" paleomagnetic pole; e.g., it determines the sense and magnitude of the intra-plate rotations as well as governs the reconstruction of South China with respect to neighboring plates. Choosing the "true" reference pole has considerable bearing when employing the small circle intersection technique to calculate the overall mean pole (Besse and Courtillot, 1991), as Yang and Besse (2001) used to construct the South China APWP. To use

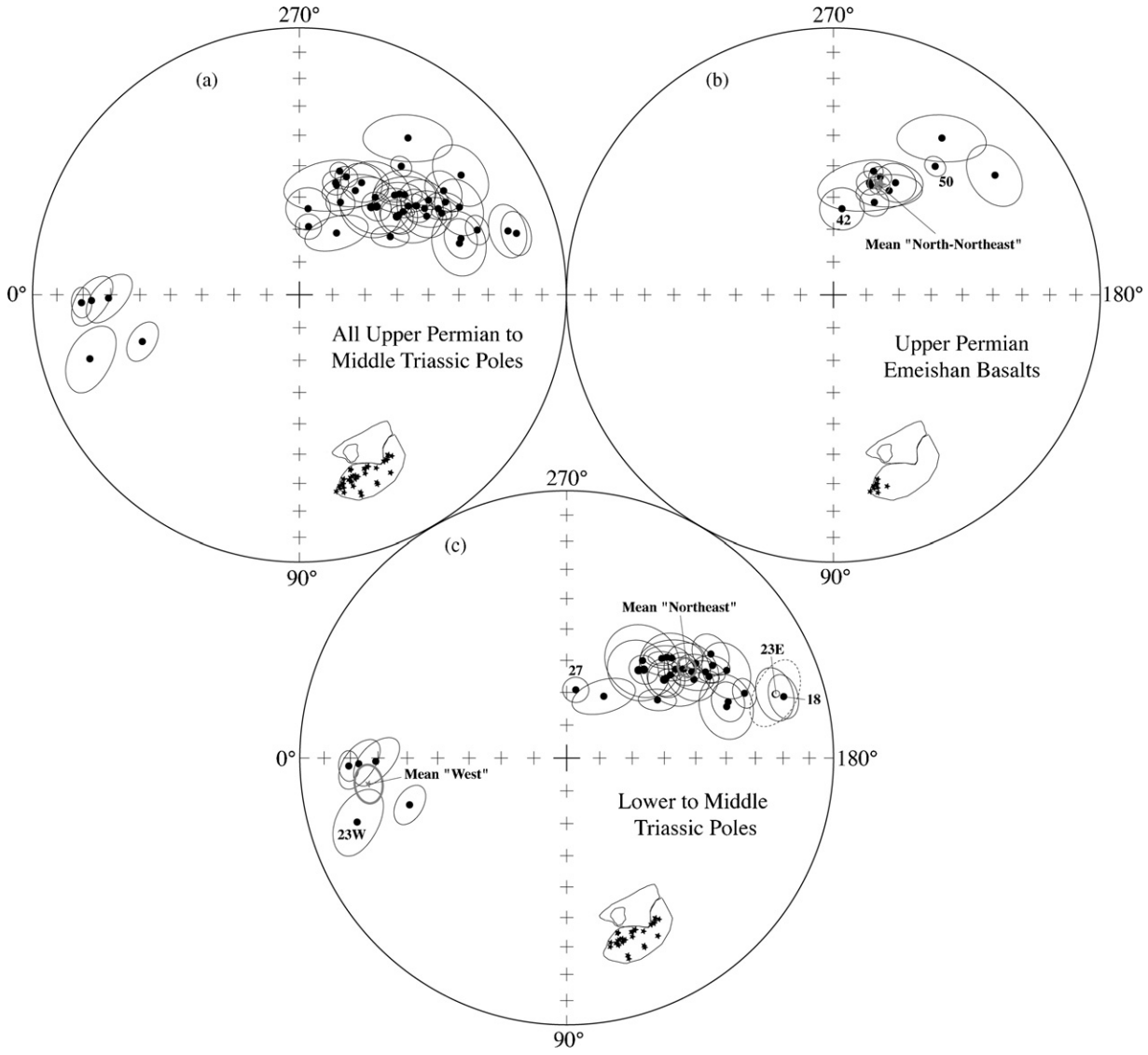


Fig. 2. Upper Permian to Middle Triassic paleomagnetic poles from the South China plate (data from Table 2). Sampling localities represented as black stars in the South China plate; some individual pole numbers and average mean poles (gray star with gray confidence ellipses) are discussed in the text: (a) all poles; (b) Upper Permian poles from the Emeishan basalts; (c) Lower to Middle Triassic poles. Individual and average mean poles indicated in (c) were used in the reconstructions shown in Fig. 3.

the small circle method, one fixes certain poles while allowing others to freely rotate about their sampling localities (i.e., move along small circle paths) toward the fixed poles. An overall mean pole is calculated using Fisher statistics by accounting for each fixed pole and for each rotated pole in the position on the small circle where the rotated pole lies closest to the group of fixed poles (McFadden and McElhinny, 1988). Thus, choosing which poles are fixed biases the outcome. This method also considerably reduces the uncertainty (radius of a cone about the mean set at 95% confidence limits, or A_{95}) compared to that of the original dataset.

Because the individual Late Permian to Middle Triassic paleomagnetic poles are so widely scattered, and because the South China plate spans about 15° in latitude and 20° in longitude (presently), there is a large enough variability in latitude and longitude that any designated reference pole influences the expected paleolatitude at each point on the plate (Fig. 3). This means that the "true" reference pole will be the one that minimizes the difference between the expected and observed paleolatitude for each point on the plate.

In this study, we attempt to better resolve the South China APWP, firstly by establishing a new method to help choose among several candidate reference poles. Secondly, we undertook a paleomagnetic sampling campaign in the Shiwandashan region of southern Guangxi Province (Figs. 1 and 4). Guangxi lies over 15° in latitude away from the North–South China plate suture zone, yet Middle Triassic rocks at two localities were found to have ENE-trending declinations, like those near the suture (Gilder et al., 1995). Tertiary, Cretaceous and Jurassic paleomagnetic data are also available for the area (Gilder et al., 1993a,b; Zhao et al., 1994). We sampled Lower Triassic carbonates and volcanic rocks to test the consistency of the rotations and the paleolatitudes from the South China plate, for our sampling locality presently lies near the southern-most point on the plate. We also collected Upper Triassic rocks to better define a "local" APWP for the Shiwandashan region in order to compare with other places from South China—notably from Sichuan Province—where several paleomagnetic studies have been carried out. This could potentially enable us to constrain the timing of the rotations in southern Guangxi Province.

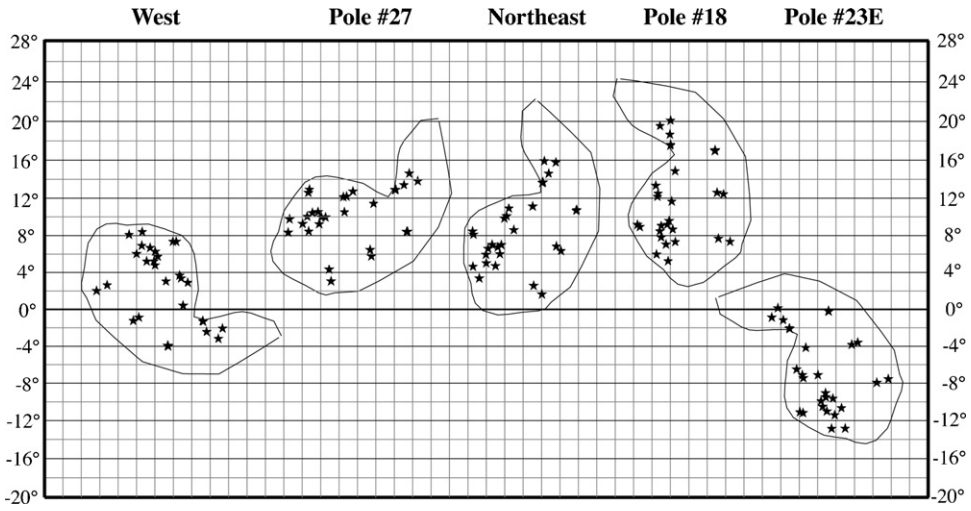


Fig. 3. Paleogeographic reconstruction of the South China plate using the individual or average mean Early to Middle Triassic poles indicated in Fig. 2c. The reconstruction with pole #23 used the clockwise-rotated (east—E) option. By using the counterclockwise-rotated option for pole #23 (Fig. 2), the reconstruction would resemble that using the "West" average pole. Each reference pole predicts a unique distribution of expected paleolatitudes for each point on the plate.

2. Local geology, sampling and laboratory treatment

We collected 52 Early Triassic samples from seven sites in the Majiaoling Formation at the Chongzuo Dam (Chongzuo County), which contains a well-exposed, ca. 500-m-thick section of gray, dolomitic limestone overlain by volcanic rocks (Fig. 4; Table 1). The carbonates are reported to contain the ammonite *Meekoceras* sp. (likely confused for *Mesohedenstroemia,* which is also Smithian in age) in the upper part and *Prionolobus* sp. in the lower part of the section, which are typical of the Lower Triassic (Guangxi Bureau of Geology and Mineral Resources, 1985, pp. 212–214; Galfetti et al., 2007). Four sites of carbonates were collected along a 30-m-long

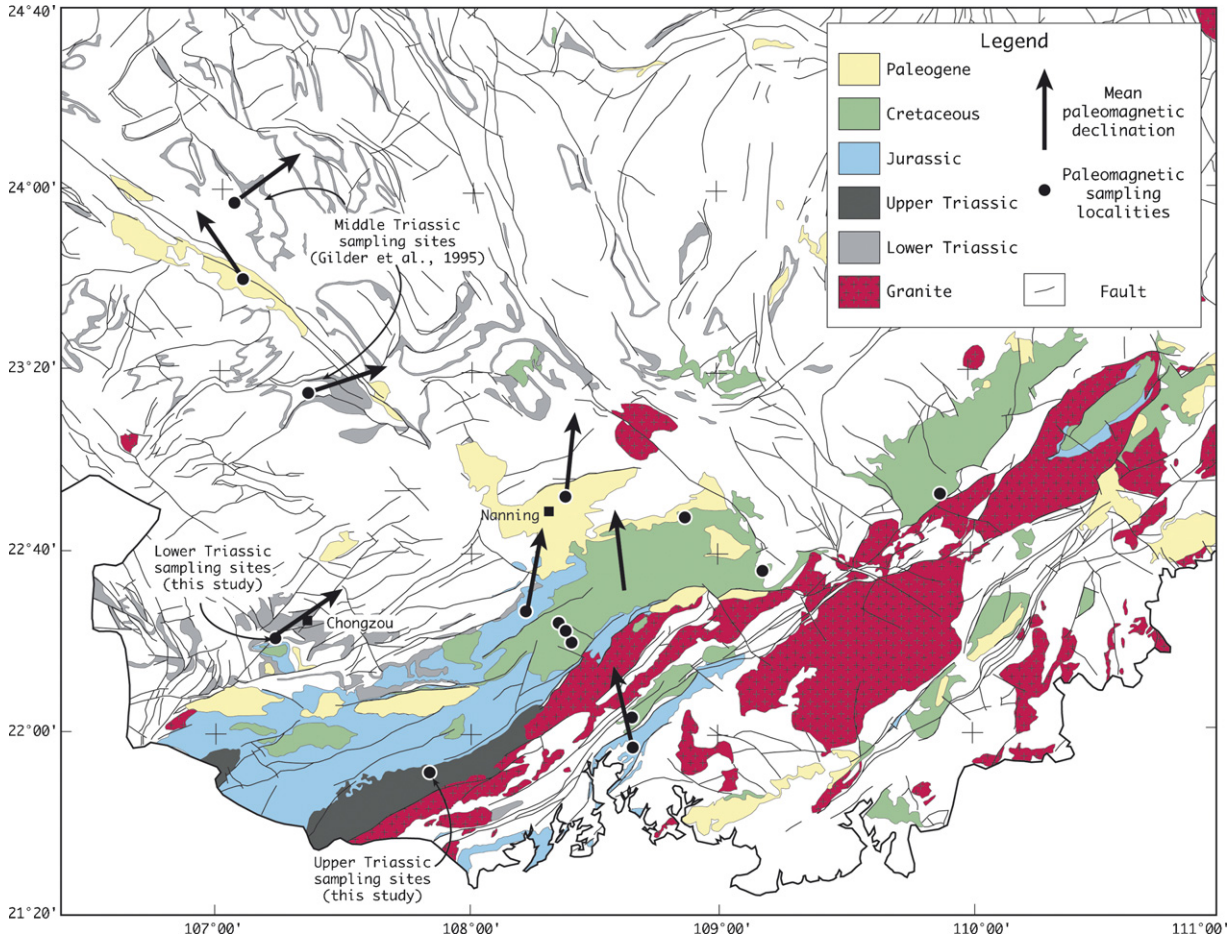


Fig. 4. Simplified geological map of southern Guangxi Province (from Guangxi Bureau of Geology and Mineral Resources, 1985). Paleogene, Cretaceous and Jurassic paleomagnetic declinations, viewed with respect to present-day north, are from Zhao et al. (1994) and Gilder et al. (1993a,b).

Table 1
Early and Late Triassic paleomagnetic data from this study

Site	Comp	n/N	D_g	I_g	D_s	I_s	k	α_{95}	Strike	Dip	Slat (°N)	Slon (°E)
Early Triassic											22°20'	107°14'
GX01	HTC	6/8	37.9	4.7	50.9	22.5	319.0	4.2	62.8	52.5	ls.: base of section	
GX02	HTC	8/8	43.1	1.4	53.1	19.0	115.5	5.6	65.8	54.0	ls.: 3 m above GX01	
GX03	HTC	8/8	45.5	5.5	57.0	20.6	91.6	6.3	68.3	48.5	ls.: 10 m above GX02	
GX04	HTC	7/8	53.4	−8.4	55.5	12.8	76.5	7.3	75.3	55.5	ls.: 15 m above GX03	
GX05	HTC	7/8	206.0	−35.5	241.0	−33.9	51.6	8.5	41.8	47.0	tuff: 150 m above GX04	
GX06	–	0/9	–	–	–	–	–	–	41.8	47.0	rhy.: 150 m above GX05	
GX07*	HTC	3/3	215.7	−13.8	227.9	−13.8	23.2	26.2	41.8	47.0	bas.: 150 m above GX06	
Mean	HTC	5/7	41.8	7.7			18.3	18.4				
Mean	HTC	5/7			55.4	21.8	91.6	8.0				
Late Triassic												
GX08	ITC	7/8	347.8	0.9	349.0	−13.8	103.9	6.0	222.8	18.0	21°50.814'	107°53.933'
GX09	ITC	8/8	347.1	−1.0	349.7	−23.0	76.2	6.4	229.8	25.0	21°50.773'	107°53.863'
GX10	ITC	8/9	350.2	8.9	350.6	−15.7	116.6	5.2	247.5	25.3	21°50.689'	107°53.564'
GX11	ITC	8/8	353.6	18.5	353.7	−19.1	160.0	4.4	254.5	38.0	21°50.440'	107°52.538'
GX12	ITC	6/8	343.4	36.5	341.8	2.8	330.5	3.7	245.1	34.0	21°52.088'	107°50.443'
GX13	ITC	8/8	2.4	36.9	354.5	14.6	134.2	4.8	232.5	27.3	21°51.992'	107°50.705'
GX14	ITC	7/8	351.7	41.4	347.8	10	208.4	4.2	245.5	32.3	21°52.117'	107°51.188'
GX15	ITC	8/8	351.5	40.8	348.6	0.4	199.7	3.9	249.8	41.0	21°51.772'	107°51.313'
GX16	ITC	7/8	349.3	41.8	348.2	4.4	251.7	3.8	254.8	37.5	21°51.763'	107°51.595'
GX17	ITC	7/8	348.0	35.3	347.8	−6.7	201.4	4.3	256.8	42.0	21°50.725'	107°51.831'
Mean	ITC	8/10	350.4	26.2			20.6	10.9				
Mean	ITC	8/10			349.1	−4.6	36.8	8.1				
GX09	HTC	7/8	267.1	−32.7	249.1	−44.9	38.6	10.8	229.8	25.0	21°50.773'	107°53.863'
GX10	HTC	8/9	80.0	35.7	61.0	37.1	103.7	5.6	247.5	25.3	21°50.689'	107°53.564'
GX11	HTC	6/8	255.7	−28.5	236.8	−22.8	92.2	7.5	254.5	38.0	21°50.440'	107°52.538'
GX17	HTC	5/8	270.5	−35.8	239.3	−34.3	84.8	9.5	256.8	42.0	21°50.725'	107°51.831'
Mean	HTC	4/10	88.2	33.3			151.1	7.5				
Mean	HTC	4/10			61.2	34.8	65.1	11.5				

Abbreviations are: Comp., magnetization component; HTC, high temperature component; ITC, intermediate temperature component; n/N, number of samples used to calculate mean direction/number of samples demagnetized; D, declination; I, inclination; g, in-situ (geographic) coordinates; s, tilt-corrected (stratigraphic) coordinates (at 100% unfolding); k, the best estimate of the precision parameter; α_{95}, the radius that the mean direction lies within 95% confidence; strike is dip direction minus 90°; Slon and Slat, longitude and latitude of site; ls., limestone; rhy., rhyolite; bas., basaltic andesite; *Not included in mean due to large α_{95}.

profile. Above the carbonates lies a thick pile of volcanic rocks, in which we collected ash-fall tuff, rhyolite and basaltic andesite. 81 samples (10 sites) of Upper Triassic red sandstones and siltstones belonging to the Fulong'ao Formation were collected along a circuitous road that transects the Shiwandashan Mountains (Fig. 4; Table 1). Abundant plant fossils constrain the Late Triassic age of these red beds (Guangxi Bureau of Geology and Mineral Resources, 1985, pp. 222–225).

Paleomagnetic cores were collected using a portable gasoline-powered drill and oriented with a magnetic compass. The absence of sun made sun compass corrections impossible to ascertain potential declination anomalies, so each core azimuth and bedding strike were corrected by −1.2° as calculated from the International Geomagnetic Reference Field (October, 2000). Cores were cut into 2.2-cm-long samples whose magnetic remanences were measured with a 2G Inc. 3-axis, DC-SQUID magnetometer installed in a magnetically shielded room at the Institut de Physique du Globe de Paris. Each sample underwent thermal demagnetization with an average of 18 demagnetization steps. Magnetization components were determined using principal component analysis (Kirschvink, 1980). When multiple components with overlapping spectra were present, we used the remagnetization great circle method of McFadden and McElhinny (1988). Site-mean directions were calculated using Fisher (1953) statistics. Data were treated using the software of Cogné (2003) and R. Enkin (unpublished).

3. Paleomagnetic results

3.1. Lower Triassic Majiaoling Formation

Stepwise demagnetization of the Early Triassic carbonates (sites GX01–GX04) typically isolated two magnetization components (Fig. 5a–d). One trends north and down (in situ coordinates) below ~250 °C that does not trend toward the origin on orthogonal vector plots. The second component has northeast-trending declinations with shallow inclinations (in situ coordinates) that trend toward the origin on orthogonal vector plots upon increasing levels of demagnetization. This higher temperature magnetization component is commonly removed by 500 °C although it sometimes persists to ~550 °C. For a few samples, magnetization intensity rapidly drops below 100 °C. These samples also have high coercivities, indicating the presence of iron hydroxide. Both the low and high temperature component directions for all carbonate samples are solely of normal polarity. Samples from site GX05 also have two magnetization components. One has north and down directions (in situ coordinates) below 200–300 °C that do not decay toward the origin. These give way to the second component that does decay toward the origin, characterized by southwesterly declinations with upward inclinations that completely demagnetize around 580 °C (Fig. 5e). Most samples from site GX06 had multiple magnetization components (Fig. 5f), yet none of the directions are consistent from one sample to another. No site-mean direction could be calculated for GX06. Magnetization directions from the three samples from site GX07 remain virtually unchanged during stepwise demagnetization from the first step to the last (Fig. 5g). All polarities are reversed. Although Table 1 lists the mean direction for the three samples, the radius that the mean direction lies within 95% confidence (α_{95}) is quite large (26.2°), so we did not consider it when applying the fold test.

The fold test based on the site-mean low temperature magnetization components is clearly negative, with the best estimate of the precision parameter (k) decreasing more than 10-fold before ($k = 369$) versus after ($k = 35$) unfolding. The overall mean in situ direction (declination, $D = 0.4°$, inclination, $I = 47.6°$, $\alpha_{95} = 4.0°$) is distinguishable at the 95% confidence level from both the present-

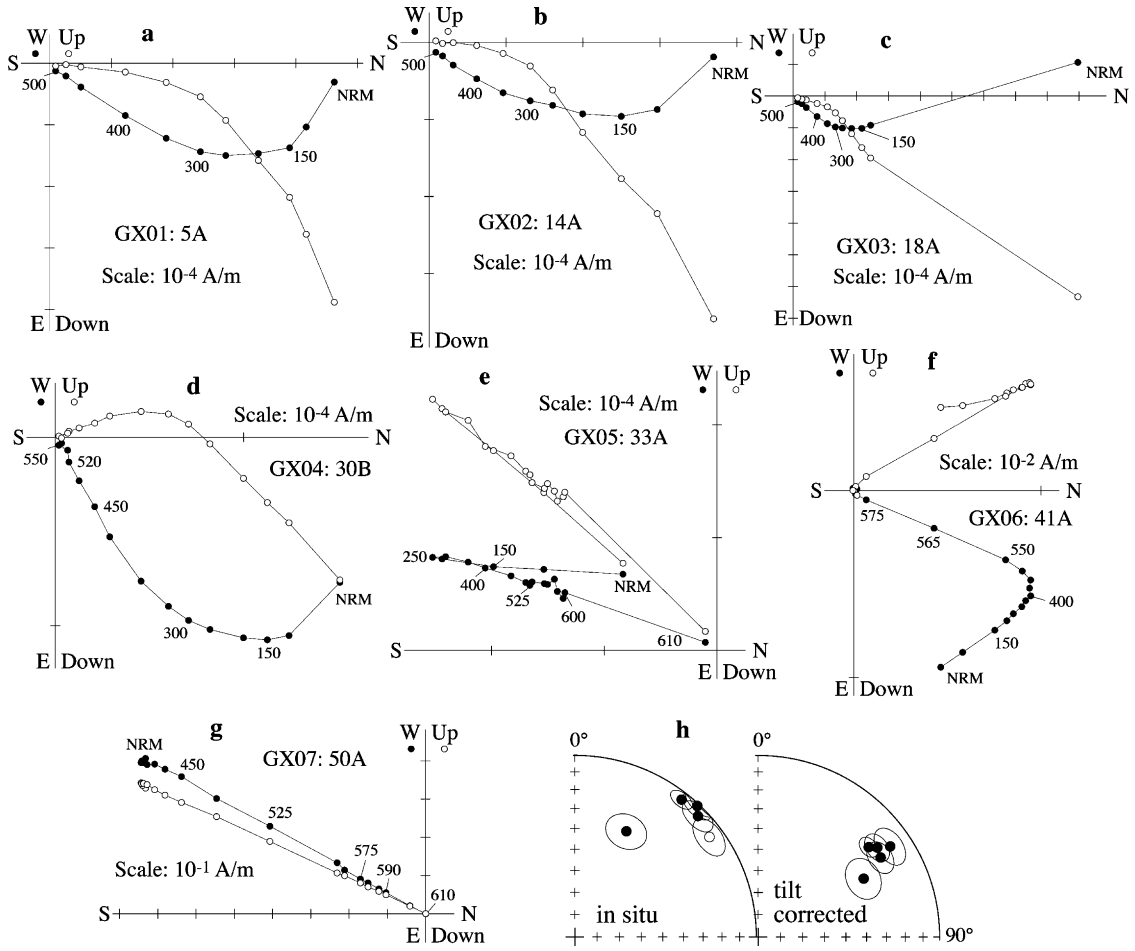


Fig. 5. Orthogonal demagnetization diagrams (in situ coordinates) of the Lower Triassic Majiaoling Formation for (a–d) dolomitic limestone, (e) ash-fall tuff, (f) rhyolite, and (g) basaltic andesite. Site and sample numbers are indicated on each figure. Temperature steps are in degrees Celsius, with NRM (natural remanent magnetization) being the room temperature step. (h) Site-mean directions with α_{95} confidence ellipses for sites GX01–GX05 (data in Table 1). All site-mean directions are shown with the same polarity for better comparison.

day field ($D = -1.2°$, $I = 31.3°$) and geocentric axial dipole ($D = 0°$, $I = 38.9°$) directions. This is in contrast to the average natural (room temperature) remanent magnetization (NRM) direction for the carbonate samples ($D = 5.7°$, $I = 36.0°$, $\alpha_{95} = 2.9°$) which lies close to the recent geocentric axial dipole direction. Given the solely normal polarities, the low temperature component could have been acquired during the Cretaceous long normal superchron, if folding took place before the Cretaceous. We suspect that folding, or perhaps a second generation of folding, of these beds took place during the Cenozoic during extrusion of Sundaland along the nearby Red River fault (Fig. 1; Gilder et al., 1995), which makes the interpretation of the age of this component difficult, as a remagnetization event would have to have taken place solely within a normal polarity chron within the Cenozoic.

Table 1 lists the site-mean directions of the high temperature magnetization components. Based on five site-mean directions, k increases 5 times from 18.3 to 91.6 upon unfolding. The direction-correlation test (Enkin, 2003) applied to the site-mean directions yields an indeterminate result with a maximum at $173 \pm 33\%$ unfolding. When applying the Watson and Enkin (1993) fold test, which allows one to assign circular standard deviations of bedding plane observations (here 8° considering higher uncertainty on the volcanic horizons), the optimum degree of untilting occurs at $129 \pm 20\%$. As the overall mean direction at 130% ($D = 62.1°$, $I = 23.3°$, $\alpha_{95} = 5.4°$) is not significantly different than at 100% ($D = 55.4°$, $I = 21.8°$, $\alpha_{95} = 8.0°$) unfolding, we assume the latter represents a pre-folding magnetization, and given the dual polarities, this magnetization component is likely a primary, Early Triassic remanence. Heat from the volcanic rocks could not have remagnetized the underlying carbonates because the polarities are different.

The magnetization direction for the Lower Triassic Majiaoling Formation is indistinguishable at 95% significance limits from the mean direction of two studies on Middle Triassic siltstone and shale located about 100 km north of Chongzuo ($D = 56.1°$, $I = 6.9°$, $\alpha_{95} = 9.8°$) (Fig. 4). This reinforces the interpretation that both results stem from primary remanent magnetizations. Although the declinations agree to within 1°, the inclinations are 15° apart. One explanation for the difference is that inclination shallowing is more prevalent in the clastic sediments than in the carbonates.

3.2. Upper Triassic Fulong'ao Formation

Most Upper Triassic red beds display fairly straightforward demagnetization trajectories (Fig. 6a–j). NRM directions roughly segregate into two populations: one that trends north and down (in situ coordinates: $D = 353.2°$, $I = 39.4°$, $\alpha_{95} = 3.0°$) that lies close to the recent geocentric axial dipole direction and another with northward declinations and shallow inclinations (in situ coordinates: $D = 342.6°$, $I = 8.4°$, $\alpha_{95} = 4.2°$) (Fig. 6k). All NRM directions cluster slightly better after ($k = 20.6$, $\alpha_{95} = 3.3°$) rather than before ($k = 15.9$, $\alpha_{95} = 3.8°$) unfolding. After the first few demagnetization steps, 75 of 81 samples have one dominant component that

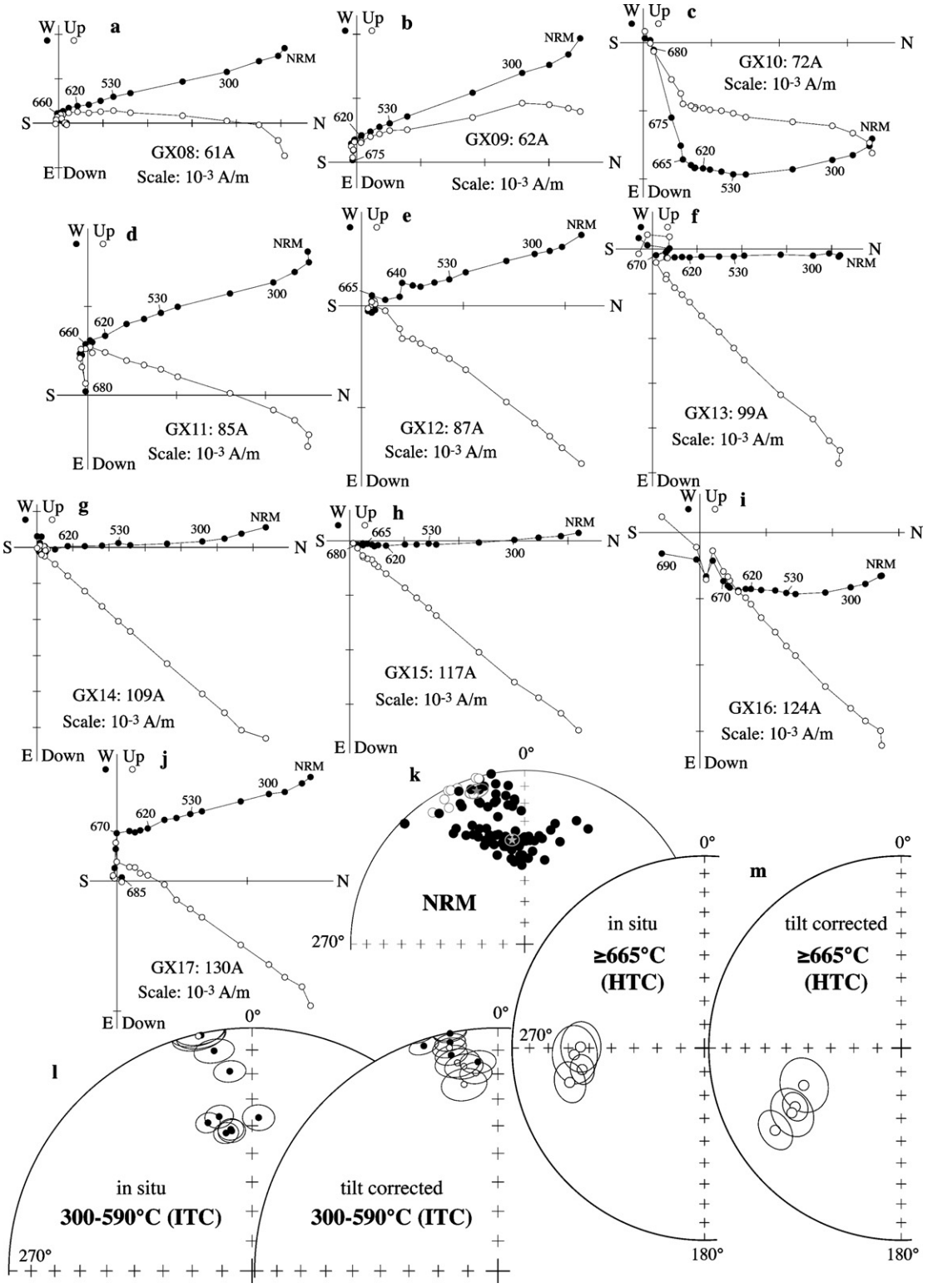


Fig. 6. Orthogonal demagnetization diagrams (in situ coordinates) of the Upper Triassic Fulong'ao Formation red beds. Site and sample numbers are indicated on each figure. Temperature steps are in degrees Celsius, with NRM (natural remanent magnetization) being the room temperature step. (k) NRM directions roughly separate into two populations whose means are indicated with gray stars and α_{95} confidence ellipses. (l) Site-mean directions of the intermediate temperature magnetization component (ITC) with α_{95} confidence ellipses (data from Table 1); all ITC directions are of normal polarity only. (m) Site-mean directions of the high temperature magnetization component (HTC) with α_{95} confidence ellipses (data from Table 1). HTC site-mean directions are shown with the same polarity for better comparison.

Table 2
Late Permian to Middle Triassic paleomagnetic poles from the South China plate

Pole #	Age	Slat	Slon	Plat	Plon	A_{95} (dp/dm)	Reference
1	Tl	28.2	103.0	56.8	212.6	3.2/5.9	Zhu et al. (1988)
2	Tl	29.6	103.4	48.8	218.7	1.8/3.6	Ma and Zhang (1989)
3	Pu–Tl	28.9	104.9	43.3	211.7	3.2/6.3	Enkin et al. (1992)
4	Tl	28.0	105.5	44.1	217.2	4.9	Yang and Besse (2001)
5	Pu–Tl	29.6	105.6	45.8	225.2	3.5/7.0	Steiner et al. (1989)
6	Tl	32.1	106.2	42.5	214.6	8.5	Bai et al. (1997)
7	Tl	29.9	106.3	38.7	211.8	3.2/6.3	Steiner et al. (1989)
8	Tl	29.9	106.3	38.5	209.8	1.7/3.3	Steiner et al. (1989)
9	Pu–Tl	32.4	106.4	47.2	226.3	2.1/4.2	Steiner et al. (1989)
10	Tl	28.6	106.9	46.3	219.2	10.9	Opdyke et al. (1986)
11	Pu–Tl	29.9	107.0	39.8	216.2	2.7/5.3	Enkin et al. (1992)
12	Tm	23.6	107.2	32.2	208.7	7.9	Gilder et al. (1995)
13	Tl	22.3	107.2	36.1	199.2	5.5	This study
14	Tl	29.3	107.3	44.3	217.4	3.5	Heller et al. (1995)
15	Tl	29.2	107.8	35.5	212.4	2.8/5.6	Chen et al. (1993)
16	Tl–Tm	29.4	110.2	37.1	197.8	9.0	Dobson and Heller (1993)
17	Tm	29.4	110.2	29.8	200.0	4.0	Huang and Opdyke (1996)
18	Tm	31.0	110.4	16.6	195.8	5.5	Huang and Opdyke (1996)
19	Tl	31.0	110.8	19.4	197.0	6.8	Tan et al. (2007a,b)
20	Tm	31.5	111.7	51.4	218.8	5.0	Su et al. (2005)
21	Tm	31.4	111.7	50.5	218.8	5.8	Huang and Opdyke (1997)
22	Tl	24.2	112.4	68.0	239.0	5.0/10.0	Tan et al. (2000)
23	Tl	24.2	112.4	−19.0	197.0	6.0/12.0	Tan et al. (2000)
24	Tl	24.9	112.4	29.2	1.0	4.7/9.3	Chen et al. (1993)
25	Tm	29.7	113.9	33.8	215.8	5.8	Huang and Opdyke (1996)
26	Tm	30.6	116.8	38.5	16.6	3.9/6.8	Gilder et al. (1999)
27	Tm	30.6	116.9	69.0	262.4	3.9	Gilder et al. (1999)
28	Tl	26.0	117.1	23.1	1.6	4.6/8.8	Dobson et al. (1999)
29	Tl	26.0	117.2	19.4	2.1	3.8	Gilder et al. (1995)
30	Tm	30.9	117.9	55.1	230.7	9.3	Su et al. (2005)
31	Tm	32.0	118.9	53.9	228.9	4.4	Su et al. (2005)
32	Tm	32.0	119.0	44.8	223.6	5.8	Opdyke et al. (1986)
33	Tl	31.0	119.8	51.8	232.1	11.2	Dobson et al. (1993)
41	Pu	26.8	101.8	25.6	216.5	8.7	Zhou et al. (1986)
42	Pu	27.6	101.9	63.5	264.3	6.6	Liu et al. (1987)
43	Pu	25.0	102.7	50.3	241.0	6.1	Van der Voo et al. (1993)
44	Pu	28.1	102.9	58.9	246.1	4.3	Liu et al. (1987)
45	Pu	26.7	102.9	53.5	241.8	5.0/10.0	Huang et al. (1986)
46	Pu	28.3	103.0	54.3	251.3	8.0/16.0	Zhou et al. (1986)
47	Pu	29.6	103.4	53.7	252.1	3.8	McElhinny et al. (1981)
48	Pu	29.6	103.4	50.7	248.3	3.3	Zhang (1984)
49	Pu	29.6	103.4	49.7	252.0	2.8	Zhao and Coe (1989)
50	Pu	29.7	103.5	38.5	231.6	3.1	Liu et al. (1987)
51	Pu	26.4	105.7	29.3	235.3	7.4/13.4	Lin et al. (1985)

Abbreviations are: Slat, Slon, Plat, Plon, the site (S) and pole (P) latitude (lat) and longitude (lon); A_{95}, half angle of cone at 95% confidence limits about pole; dp/dm, axes of ellipse at 95% confidence limits about pole; Tm, Middle Triassic; Tl, Lower Triassic; Pu, Upper Permian. The following poles were not included due to limited number of samples or high uncertainties: Pu pole in Chan et al. (1984); Tl pole in Chen et al. (1993); two Pu poles in Huang et al. (1986); one Pu pole in Van der Voo et al. (1993). Pole numbers are classed by Slon.

persists from 300 to ~590 °C and sometimes until the highest applied temperature steps (680–690 °C). Several samples have linear magnetization directions that often do not trend toward the origin in the 300–650 °C range, but the directions become erratic above 650 °C (Fig. 6f and i). Stepwise demagnetization above 650 °C occasionally yields a component with east and down or west and up directions that decay toward the origin on orthogonal plots (Fig. 6b–d and j) until complete demagnetization at 680–690 °C. Four sites have samples with this high temperature magnetization component (GX09–GX11 and GX17).

When applying best-fit line segments, we defined an intermediate temperature magnetization component (ITC) between 300 and 590 °C that was not anchored to the origin. This yielded site-mean directions with $k > 100$ and $\alpha_{95} < 5^\circ$ in most cases (Table 1; Fig. 6l). Every direction is solely of normal polarity. The direction-correlation test applied to the site-mean directions is indeterminate at 135 ± 104% unfolding. When applying the Watson and Enkin (1993) fold test with 5° circular standard deviations of bedding plane observations, the optimum degree of untilting occurs at 116 ± 32%, suggesting a pre-folding magnetic remanence. Site-mean directions of the high temperature magnetization component (HTC) were determined with a combination of best-fit line segments and remagnetization great circles (Fig. 6m; Table 1). High temperature magnetization components from sites GX09, GX11 and GX17 were of reverse polarity while those from GX10 were of normal polarity. The direction-correlation test applied to the site-mean directions is again indeterminate at 8 ± 209% unfolding. The Watson and Enkin (1993) fold test shows that the optimum degree of untilting occurs at −8 ± 42%, suggesting a post-folding remanence for the HTC.

It is difficult to make sense of the magnetization components in the Upper Triassic red beds. The >1000-m-thick section was sampled fairly uniformly (Table 1); we do not suspect duplication of the sequence. Each site represents about 10 m in thickness with cores spaced roughly a meter apart. Although the clustering of the ITC directions improves after unfolding, the solely normal polarities within such a thick section seems unlikely given the fairly high reversal frequency in the Upper Triassic (Gallet et al., 2007). Moreover, the paleolatitude of the ITC (−2.4 ± 3.5°) places the Shiwandashan basin farther south than the paleolatitudes of nearby

Table 3
Candidate reference poles for South China

Pole	Plat	Plon	A_{95}
Early to Middle Triassic			
West	26.0	7.5	10.5
27	69.0	262.4	3.9
Northeast	43.0	217.0	2.8
18	16.6	195.8	5.5
23E	−19.0	197.0	12.0
Late Permian			
42	63.5	264.3	6.6
NNE	53.1	247.5	3.2
50	38.5	231.6	3.1

Abbreviations are: Plat, Plon, the pole (P) latitude (lat) and longitude (lon); A_{95}, half angle of cone at 95% confidence limits about pole.

Lower and Middle Triassic rocks (e.g., $11.4 \pm 4.3^\circ$ and $3.4 \pm 6.1^\circ$). Given that the paleolatitudes of nearby Early Jurassic rocks are $12.5 \pm 5.0^\circ$ and $16.7 \pm 11.0^\circ$ (Gilder et al., 1993b) implies that Shiwandashan would have moved ~13° northwards in ca. 20 million years if the ITC were a primary magnetization.

The HTC direction possesses an equally difficult problem to solve. In this case, both polarities are represented, but the fold test is negative. The in situ mean direction of the HTC ($D = 88.2^\circ$, $I = 33.3^\circ$, $\alpha_{95} = 7.5^\circ$) is not unlike the tilt corrected directions from the nearby Lower and Middle Triassic rocks. This could suggest the Lower and Middle Triassic rocks from Guangxi rotated after the Upper Triassic but before the Lower Jurassic because declinations in the Lower Jurassic red beds are northerly. This could potentially indicate important tectonism around the Late Triassic–Early Jurassic boundary; however, such a drastic interpretation requires more convincing evidence. It is premature to properly assess the Upper Triassic paleomagnetic data before detailed rock magnetic experiments and further sampling are carried out.

3.3. Optimization of apparent polar wander paths

An important goal of our work is to establish a method to determine the "true" reference pole for a plate whose individual poles are not Fisherian distributed, but instead are streaked out about a swath, presumably due to vertical axis block rotations. The small circle intersection technique is well suited to solve this problem if one knew which pole, or group of poles, best approximated the "true" reference pole. This is unknown for the South China plate (Table 2). However, one can calculate the difference between the expected paleolatitude from a reference pole and the observed paleolatitude for each sampling locality. Remembering that the "true" reference pole will correctly predict the paleolatitude for each site on the plate, the average difference between the expected and observed paleolatitude will be closest to zero for the best-fitting reference pole. We thus chose five potential reference poles for the Early to Middle Triassic (Fig. 2, Table 3): one called "West", which is the Fisherian average of the five poles that lie to the west of South China; one is pole #27, which represents one end-member of the main group of poles; one called "Northeast" (or NE) which is the Fisherian average of the poles from Sichuan Province; one is pole #18, which represents the other end-member of the Northeast poles; and the fifth is pole #23E, which represents the eastern-most end-member of all the poles. Three potential reference poles were chosen for the Upper Permian (Fig. 2, Table 3): a Fisherian average of the main group of poles, called "north-northeast" (or NNE) and its two end-member poles #42 and #50.

Fig. 7 shows the average paleolatitude difference and associated standard deviation for each candidate reference pole. The Late Permian and Early to Middle Triassic populations are quite different in that the former come exclusively from the Emeishan flood basalts whereas the latter come from carbonates and clastic sedimentary rocks. Pole #23E and #42 predict paleolatitudes one standard deviation away from 0 and can be dismissed as potential reference poles. To better distinguish between the remaining cases, we identified the paleolatitude differences of the individual poles lying one standard deviation from the mean of every reference pole. These outlier poles obviously enlarge the standard deviation, and equally importantly, they may bias the distributions as seen in histograms discussed below. Four such poles exist for the Early to Middle Triassic (pole numbers 16, 24, 28 and 29) and one for the Late Permian (pole #51). Pole #16 was derived from the same locality as pole #17; however, the former was based on a study at the sample level (29 samples) whereas the latter at the site level (20 sites) with four times more samples (105 samples). Pole #24 also comes from a

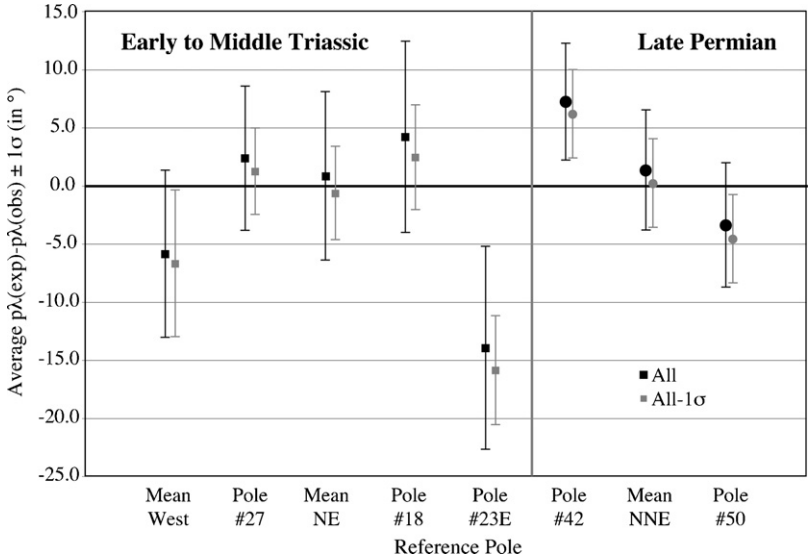


Fig. 7. Average and standard deviation of the expected paleolatitude [pλ(exp)] minus the observed paleolatitude [pλ(obs)] for 33 Early to Middle Triassic (left) and 11 Late Permian (right) paleomagnetic poles from the South China plate (black points). Five and three potential (candidate) reference poles were chosen for the Early to Middle Triassic and Late Permian data sets, respectively. When individual poles lay one standard deviation away from the average of each studied reference pole, the outliers (four poles for the Early to Middle Triassic and one for the Late Permian) were omitted and the average and standard deviation was recalculated (gray points).

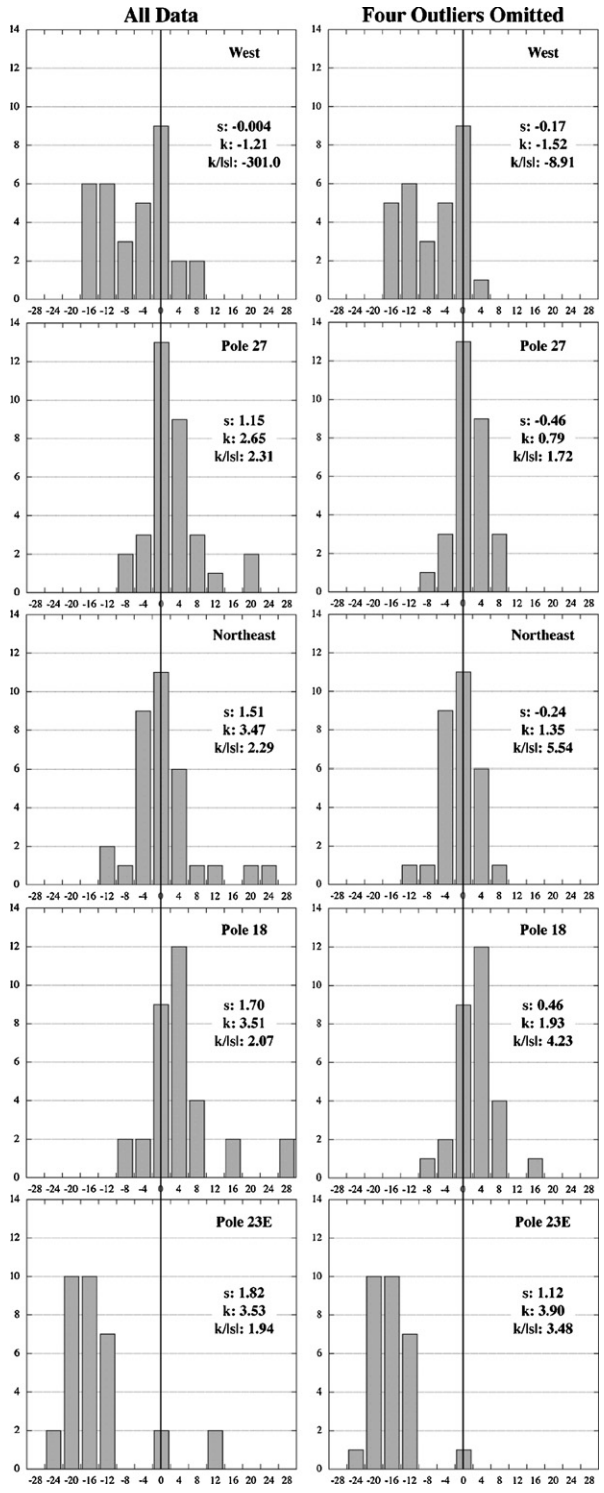


Fig. 8. Histograms of the expected paleolatitude minus the observed paleolatitude calculated from Early to Middle Triassic paleomagnetic poles from the South China plate. Five candidate reference poles were chosen for this example. The column on the left is for all 30 poles. The column on the right omits four poles that lie one standard deviation away from the average of each of the five candidate reference poles. Skewness (s) and kurtosis (k) calculated with Microsoft Excel version 11.3.7. $k/|s|$ is the ratio of the kurtosis to the absolute value of the skewness. The best candidate reference pole will have high positive $k/|s|$ with peaks centered on 0.

limited sample population (14 samples), never having been subjected to a field test. Pole #28 and #29 come from Fujian Province, which could represent an exotic terrain, not belonging to the South China plate in the Triassic (Gilder et al., 1995; Dobson et al., 1999). The outlier pole (#51) from the Emeishan basalts has the second highest A_{95} of all the Late Permian poles. It could depict anomalous field behavior (secular variation not averaged).

The potential "true" reference pole becomes more clearly identified after omitting the outlying poles and recalculating the averages and standard deviations (gray bars in Fig. 7). Only one (the NNE pole) of the three Late Permian reference poles we considered matches the observed paleolatitudes within one standard deviation. We regard it as the Late Permian reference pole for the South China plate. Omitting the outlying Early to Middle Triassic poles naturally lowers the standard deviations and allows us to omit "West" as a candidate reference pole. To better analyze the three remaining potential reference poles, we made histograms for each of the five considered reference poles by considering all data ($N = 33$) and for the subset of data that eliminates the four outliers ($N = 29$) (Fig. 8).

Besides having paleolatitude differences averaging 0, the best candidate reference pole will have histogram populations that display low skewness and high kurtosis. Symmetrically distributed data has a skewness around zero. Negative skewness indicates data that are skewed to the left and positive skewness values indicate data that are skewed to the right. A Gaussian (normal) distribution has a kurtosis of zero; positive kurtosis indicates a pointed or peaked distribution while negative kurtosis indicates a flat distribution. Although methods for calculating skewness and kurtosis vary, we found that their relative values remain relatively constant. As the absolute sign of the skewness is unimportant in the present analysis (although it could potentially be used to discriminate inclination shallowing), the "true" reference pole will have the highest positive ratio of kurtosis to the absolute value of skewness. These values are provided in Fig. 8. Viewed with all data, pole #27, Northeast and #18 remain roughly equally plausible. For the subset of data that omitted the four outliers, "Northeast" has the highest positive ratio of kurtosis to the absolute value of skewness and is best centered on 0, which together point toward it as the best candidate pole. This would again confirm the assumption that the average of the poles from Sichuan Province is most representative for the South China plate.

Here we assume that no significant apparent polar wander of the South China plate occurred during the Early to Middle Triassic. Motion during this interval could skew the data. To test this, we broke the data into two subsets, only Early Triassic and only Middle Triassic, and then repeated the exercise despite the low number of data (again excluding outliers). Potential Early Triassic reference poles considered were numbers 7, 13, 22 and "Northeast". Poles 13 and 22 lay one standard deviation away from 0. The ratio of kurtosis to the absolute value of skewness is nearly two times greater for "Northeast" than pole #7, which substantiates "Northeast" as the Early Triassic reference pole for South China. For the Middle Triassic, we chose five potential reference poles: 12, 27, 31, 32 and "Northeast". The analysis identifies pole #27 as the most likely; however, we caution against using it as the Middle Triassic reference pole until Middle Triassic paleomagnetic data become available from Sichuan Province.

4. Discussion and conclusions

We present a method to identify the "true" reference pole from a distribution of poles that were potentially dispersed from their original position due to vertical axis block rotation, translation, etc.

As with any statistical method, the outcome is highly dependent on the input parameters. One shortcoming is the way the individual poles were averaged. An analysis based on site-mean poles rather than study-mean poles would be preferable because the number of data would significantly increase and the weighting of each pole would be more equal. Another potential problem arises with how the magnetization acquisition process might influence the pole position. For the Late Permian case studied here, the "true" reference pole was quickly obtained based on the average and standard deviation of the various populations, despite the fact that the study localities are much less spatially distributed than for the Triassic poles. The likely reason for this is that all of the Late Permian poles come from one, coeval and lithologically homogeneous rock type. Thus, the magnetic recording process is the same in each study and the ambiguity surrounding the age of the rock relatively low. On the other hand, the Triassic data come from a wider variety of rock types whose ages are less coeval. Thus the magnetic recording process is not constant from one datum to the next and aberrations such as syn-sedimentary or compaction-induced inclination shallowing should be considered. Separating the data into subsets based on lithology might explain the skewness, where clastic sediments might have systematically lower paleolatitudes.

Despite the drawbacks, this relatively simple method can potentially shed light on APWP debates on other plates. One example is the controversy surrounding the Late Triassic to Early Jurassic part of the North American APWP (e.g., Gordon et al., 1984; Kent and Tauxe, 2005; Tan et al., 2007b; Kent and Olsen, 2008). Another application would be to test the transferred master curve of synthetic APWPs on the individual studies from each plate (Besse and Courtillot, 2002).

Part of our original objectives was to establish a local APWP for the Shiwandashan region. The consistency of the Early Triassic result from the Majiaoling Formation with Early to Middle Triassic paleomagnetic data from elsewhere in Guangxi and with the South China plate lends credence that Guangxi was part of the South China plate since at least the earliest Mesozoic. On the other hand, Late Triassic paleomagnetic data are needed from the South China plate to better understand the kinematic history of the North-South China collision and the local deformation history of the Shiwandashan area. Better understanding of the complex magnetization of the Fulong'ao Formation red beds would be welcome future work.

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